

A decorative graphic on the left side of the slide. It consists of a blue parallelogram and a light green parallelogram, both tilted at an angle. The blue shape is in the foreground, and the green shape is partially behind it. They are set against a dark blue background with faint, lighter blue diagonal stripes.

Bastian Mini Project 2

Dataset explanation

My dataset contains the house price of unit area in Taiwan from 2012 to 2013
With features based on distance from certain features as well as dates like transaction date and house age.

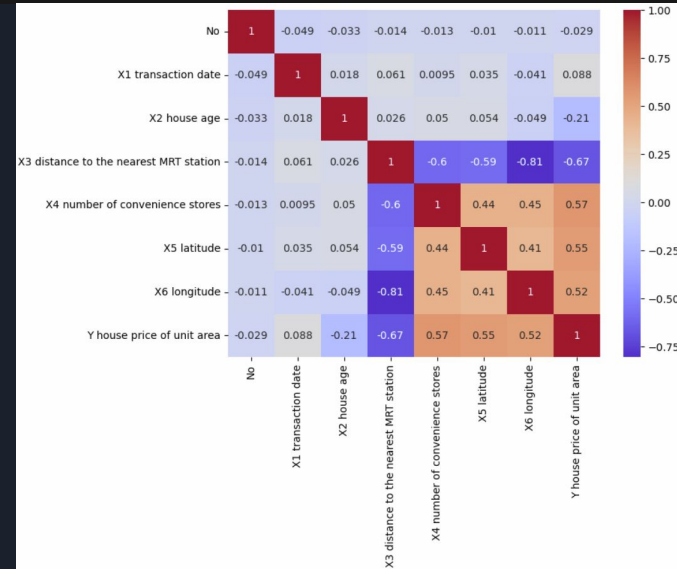
Variables Table

Variable Name	Role	Type	Description	Units	Missing Values
No	ID	Integer			no
X1 transaction date	Feature	Continuous	for example, 2013.250=2013 March, 2013.500=2013 June, etc.		no
X2 house age	Feature	Continuous		year	no
X3 distance to the nearest MRT station	Feature	Continuous		meter	no
X4 number of convenience stores	Feature	Integer	number of convenience stores in the living circle on foot	integer	no
X5 latitude	Feature	Continuous	geographic coordinate, latitude	degree	no
X6 longitude	Feature	Continuous	geographic coordinate, longitude	degree	no
Y house price of unit area	Target	Continuous	10000 New Taiwan Dollar/Ping, where Ping is a local unit, 1 Ping = 3.3 meter squared	10000 New Taiwan Dollar/Ping	no

Data exploration

When looking at the dataset there were no missing variables that I had to deal with. I then proceeded to look at the some of the datasets info by using .describe and .corr due to No seemingly just basically being an id that won't give any actual info I proceeded to drop it.

	No	X1 transaction date	X2 house age	X3 distance to the nearest MRT station	X4 number of convenience stores	X5 latitude	X6 longitude	Y house price of unit area
count	414.000000	414.000000	414.000000	414.000000	414.000000	414.000000	414.000000	414.000000
mean	207.500000	2013.148953	17.712560	1083.885689	4.094203	24.969030	121.533361	37.980193
std	119.655756	0.281995	11.392485	1262.109595	2.945562	0.012410	0.015347	13.606488
min	1.000000	2012.666667	0.000000	23.382840	0.000000	24.932070	121.473530	7.600000
25%	104.250000	2012.916667	9.025000	289.324800	1.000000	24.963000	121.528085	27.700000
50%	207.500000	2013.166667	16.100000	492.231300	4.000000	24.971100	121.538630	38.450000
75%	310.750000	2013.416667	28.150000	1454.279000	6.000000	24.977455	121.543305	46.600000
max	414.000000	2013.583333	43.800000	6488.021000	10.000000	25.014590	121.566270	117.500000



There seems to be a negative correlation with distance to station and house price most likely due to houses being much smaller in busier places.

Long and lat also have decent correlation I assume cause there is a richer location farther north of Taiwan.

Transaction date doesnt have much say cause the timeline is short 2012 to 2013



Model and feature selection

For this mini project I have selected to use Linear Regression RidgeCV and Lasso for the models to use meanwhile I have experimented with a selection of features to see if which would improve my model.

Some examples of feature selections that I made are:

All features , no lat and long , specifically no long as it seemed to have very little effect

No transaction date.

Afterwards cause of the lack of major differences I tried using forward feature selection.



Some observations

Lack of longitude results seem to be slightly better than with it but to such a minor degree that it's better to say longitude doesn't effect the model at all .

Meanwhile having no coordinates at all makes the model slightly worse meaning that the price difference depending on location is mainly focused on lat.

Also did no transaction date model was slightly worse.

Will say that when I multiplied y by 10000 (cause default table has $1 = 10000$ ping) it increased the accuracy for lasso in particular this must mean that Lasso doesn't help since the increase in scale makes it just become linear and breaks the penalty search from lasso

Selection

After using forward feature selection I have decided to use the Ridge model with all features.

For some reason forward feature selection seemed to miss this option and proceeded to not add longitude for a slightly worse score (maybe something to do with the order the features were added.)

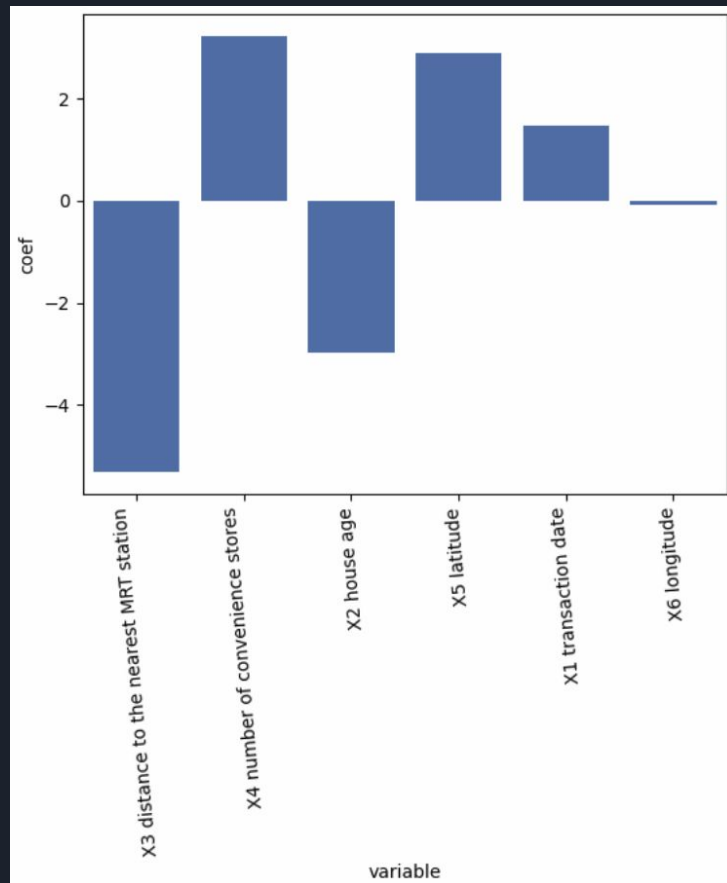
The model stats:

RMSE: 7.286

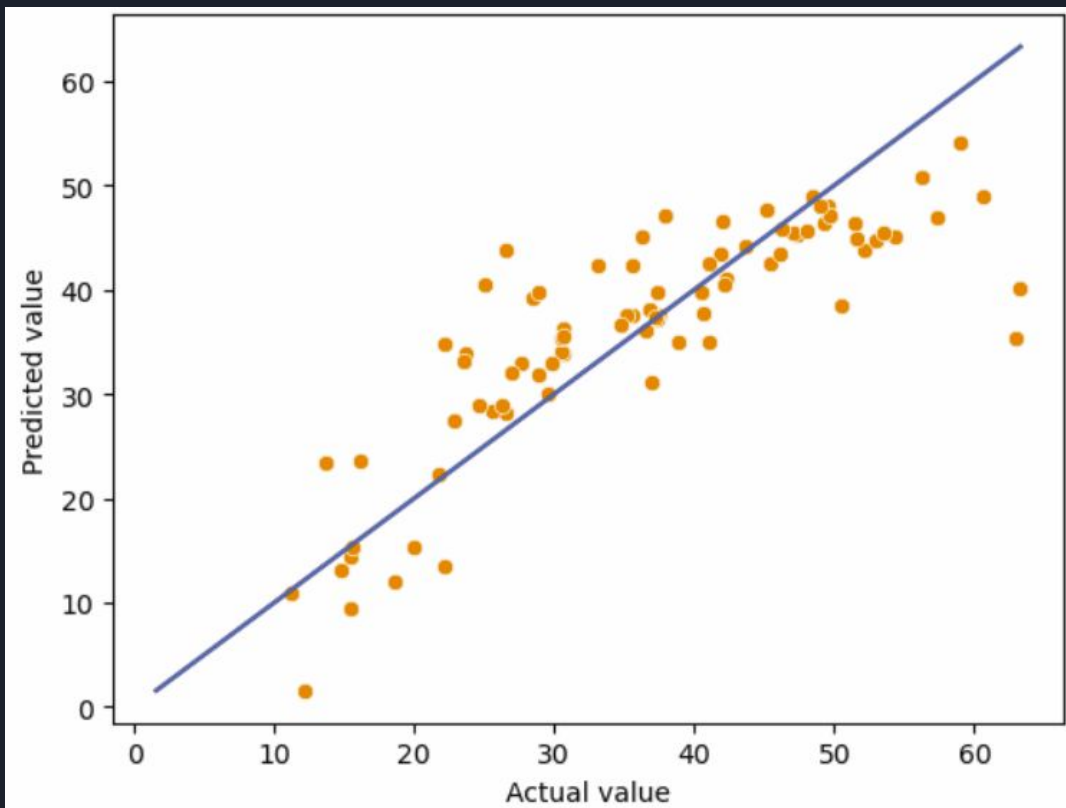
MAE: 5.295

R^2 : 0.684

Adjusted R^2 : 0.679



Prediction compared to results





Some following notes

When running train test split multiple times without random state the difference on the r^2 score seemed to range from 0.3 to 0.7 this most likely means that the predictor values don't have enough correlation to consistently make accurate predictions or there just wasn't enough data .